

Benchmarking stimulus delivery and equilibration in cell-matched neuronal model replay

Original research article

Running title: Stimulus-delivery and equilibration benchmarks

Abstract

Deterministic reruns of neuronal simulations can reproduce archived outputs while still misrepresenting the commanded stimulus or beginning from a drifting model state. We evaluated these risks in six Allen Cell Types specimens from six mice: one method-development specimen and five prespecified evaluation specimens. Twenty-four protocol-specific recording selections produced 72 analysis cases under two generalized leaky integrate-and-fire (GLIF) conditions and one perisomatic condition. Three outcomes produced 216 eligibility records: 122 calculated values and 94 reason-coded non-evaluable outcomes. Model-construction verification mapped all 158 fit selectors to their intended constructed-section classes, and recording metadata established file-specific SI units for all six NWB files. We then evaluated five current-playback methods using five synthetic benchmark waveforms at the original 0.025 ms solver step. None met all transition, pulse-width, amplitude, and charge criteria, so no candidate was applied to the 24 perisomatic cases. Separately, all 24 original pre-stimulus periods showed drift under at least one prespecified criterion; all extended-equilibration runs converged at 3.0 s, and 20 of 24 cases met at least one material-change criterion across all six specimens. These results show that verified input files and deterministic execution do not establish stimulus-delivery fidelity or initialization-insensitive behavior. In a separately labeled post hoc reference implementation, fixed-step boundary sampling at 0.005 and 0.0025 ms and event-timed piecewise-constant delivery met all five waveform criteria. The study provides benchmark waveforms, equilibration diagnostics, outcome-eligibility rules, and machine-readable reports for case-level validation; the six-cell design does not support population inference, model-class ranking, or broad biological validation.

Keywords: cellular electrophysiology; computational neuroscience; GLIF; NEURON; stimulus delivery; equilibration; reproducibility

Introduction

Deterministic reruns are often treated as evidence that a neuronal simulation was reproduced correctly. They establish repeatability of a particular implementation, but not necessarily numerical fidelity to the intended experiment. A high-rate current command can contain several transitions within one solver interval; sampling that command only at solver boundaries can alter transition times, pulse widths, amplitudes, or delivered charge even when every input file is correct and every rerun is byte-identical.

Initialization creates a separate numerical risk. Conductance-based models may retain voltage, gating-state, or membrane-current drift after their nominal initialization step, and extending the unstimulated period can change subsequent spikes or steady-state voltage. Input delivery and model-state stability must therefore be tested independently. The NEURON environment exposes explicit choices for fixed-step integration and time-vector playback [3,4], making it possible to benchmark stimulus delivery and equilibration before physiological interpretation.

The Allen Cell Types resource makes cell-matched electrophysiology, morphologies, and model resources available at specimen-level identifiers [5]. This enables specimen-level provenance, but it does not eliminate the distinction between source identity, successful execution, and scientific interpretation. The Neurodata Without Borders (NWB) format likewise supports structured provenance and unit metadata, while legacy files may still require file-specific interpretation [6].

Reproducible computational neuroscience also depends on transparent implementations, versioned software, model descriptions, and executable archives [10–12]. Cross-simulator work has shown that nominally equivalent neuronal models may converge only at sufficiently fine spatial and temporal discretization [11]. This observation motivates direct numerical tests of the command waveform and initialized state rather than relying on model-file identity alone.

Here we present a validation workflow comprising source and model provenance, model-construction verification, stimulus-delivery benchmarks, initialization and equilibration diagnostics, deterministic rerun checks, and explicit outcome eligibility. We applied it to one method-development specimen and five evaluation specimens, 24 prespecified recording selections, three simulation conditions, and three outcomes. The objective was to determine what the case-level results support once numerical fidelity and model-state stability are evaluated directly.

Results

A prespecified six-cell design produced 72 condition-specific cases

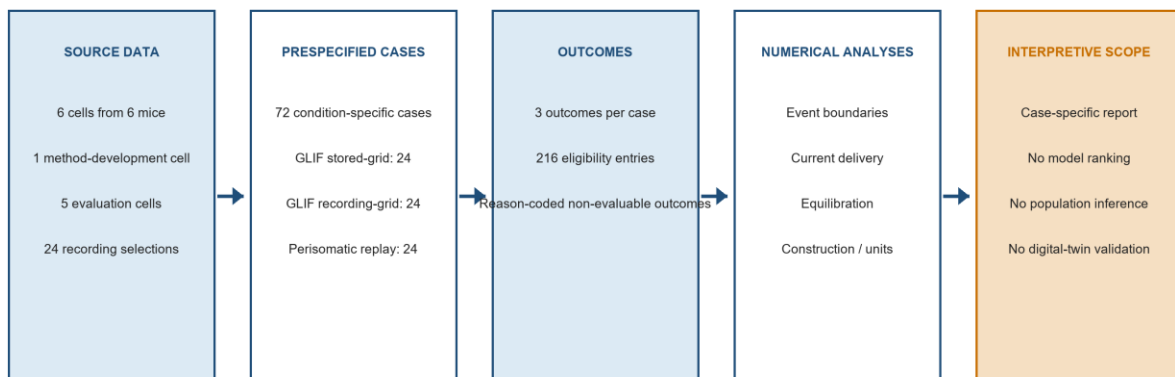
The cohort comprised six mouse visual-cortex cells, each mapped through the official Allen API to a distinct donor (Table 1). The method-development cell, specimen 314822529, was an Rorb-line layer-5 VISp cell. The five evaluation cells included one Sst-line and one Pvalb-line cell and three Htr3a-line cells across VISp layers 1, 2/3, and 6a. These source descriptors document composition; the small, unbalanced operational cohort does not support cell-type, layer, sex, or population comparisons.

The method-development specimen contributed four previously characterized raw-label selections and remained analytically separate. Each evaluation specimen contributed Long Square first-spiking, Short Square first-spiking, Short Square triple first-spiking, and Long Square prethreshold-candidate selections. The 24 selections were each represented under GLIF stored-grid, GLIF recording-grid, and perisomatic recording-grid playback, yielding 72 cases (Figure 1). Exposure of these exact sweeps during model fitting or evaluation could not be established, so no result is described as held-out or out-of-sample.

Table 1. Cohort, source animals, and analytical strata. Ages are postnatal days. Detailed specimen, model, recording, and role metadata are provided in Supplementary Table S1.

Specimen	Stratum	Donor	Sex	Age	Genotype	Hemisphere	Source structure
314822529	method development	313403626	male	P53	Rorb-IRES2-Cre/wt; Ai14(RCL-tdT)/wt	left	VISp5
324493977	evaluation	322489543	male	P54	Sst-IRES-Cre/wt; Ai14(RCL-tdT)/wt	right	VISp2/3
473943881	evaluation	472963252	female	P63	Htr3a-Cre_NO152/wt; Ai14(RCL-tdT)/wt	left	VISp2/3
471087830	evaluation	469953343	male	P55	Pvalb-IRES-Cre/wt; Ai14(RCL-tdT)/wt	left	VISp6a
475585413	evaluation	475544273	male	P59	Htr3a-Cre_NO152/wt; Ai14(RCL-tdT)/wt	right	VISp1
464188580	evaluation	418778036	female	P59	Htr3a-Cre_NO152/wt; Ai14(RCL-tdT)/wt	left	VISp2/3

Six-cell technical study: design and interpretive scope



The method-development cell is shown separately; sweeps and cases are nested technical observations.

No current-delivery candidate met all criteria; the equilibration sensitivity is reported separately.

Figure 1. Study design and interpretive scope. Source records, prespecified cases, outcomes, numerical evaluations, and interpretive limits are shown in sequence. The method-development cell remained separate, and neither cases nor sweeps were treated as biological replicates.

Outcome eligibility was retained instead of converted to failure scores

Exactly three outcomes were prespecified: absolute stimulus-window spike-count error, conditional absolute first-spike-latency error, and Long Square absolute steady-state delta-voltage error. The spike-count outcome was

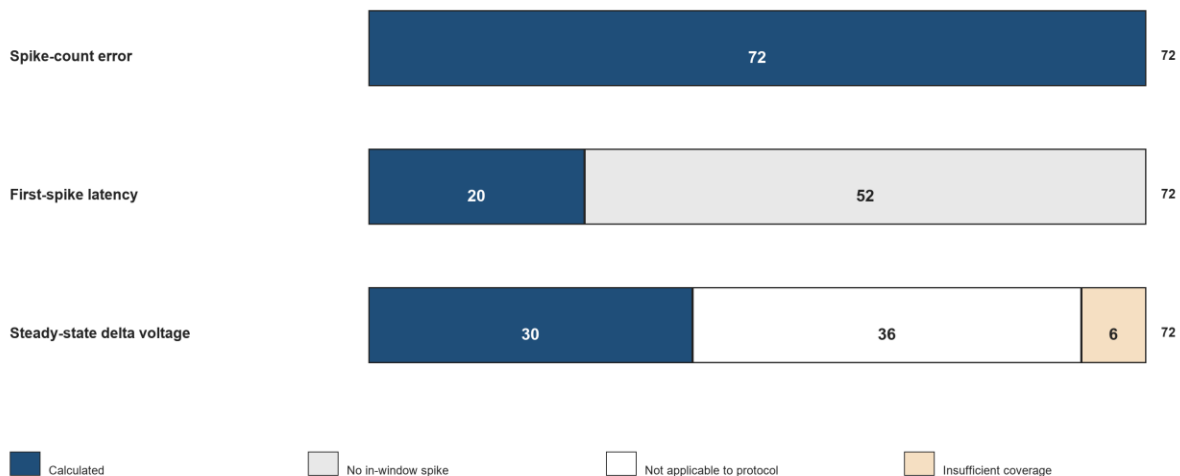
calculated in all 72 cases. Latency was calculated only when both the recording and model contained a retained in-window spike, producing 20 calculated values and 52 spike-absence non-evaluable outcomes. Steady-state delta voltage required an eligible long-duration protocol—**Long Square**, plus the method-development **Square - 2s Suprathreshold** label—and at least 0.80 usable coverage in each relevant recording and model window; 30 values were calculated, while 36 short-stimulus and six coverage-induced cases remained null (Table 2; Figure 2). Across all three outcomes, 122/216 entries were calculated and 94/216 were reason-coded as non-evaluable. No unavailable value was replaced by zero, a penalty, or a performance rank.

Table 2. Prespecified metric applicability.

Metric	Calculated	Non-evaluable	Total
Absolute spike-count error	72	0	72
Conditional absolute first-spike-latency error	20	52	72
Long Square absolute steady-state delta-voltage error	30	42	72
All metric states	122	94	216

Outcome eligibility across 72 prespecified cases

Each outcome retains calculated values and explicit reasons for non-evaluability.



Totals: 122 calculated outcomes and 94 non-evaluable outcomes (216 states).

Figure 2. Outcome eligibility across 72 prespecified cases. Stacked bars distinguish calculated values from non-evaluable outcomes caused by spike absence, protocol inapplicability, or insufficient coverage. Numbers inside the bars are case counts. The complete case-by-outcome matrix is Supplementary Figure S1 and Supplementary Table S3.

Case-specific GLIF outputs varied by cell and protocol; no condition was preferred

The two GLIF conditions differed jointly in input grid and model step: 20 kHz stored-grid execution used $dt = 50 \mu s$, whereas 200 kHz recording-grid execution used $dt = 5 \mu s$. They were therefore retained as separate technical conditions. Their contrast is a coupled input-grid and model-dt sensitivity.

Figure 3 displays raw stimulus-window spike counts for all 20 evaluation roles. For the five strict Short Square first-spiking roles, the recording and both GLIF conditions each contained zero in-window spikes. In four Long Square first-spiking roles, both GLIF conditions contained zero spikes when the corresponding recordings contained 1, 2, 1, and 4 spikes for specimens 324493977, 473943881, 475585413, and 464188580, respectively. Specimen 471087830 behaved differently: its Long Square first-spiking recording contained one spike, while the stored- and recording-grid GLIF conditions contained 71 and 79; its prethreshold recording contained zero, while the corresponding model conditions contained 57 and 62. Those two roles also failed the steady-state model-coverage criterion, so their delta-voltage outcomes remained non-evaluable rather than being forced into the steady-state comparison.

For the Short Square triple roles, recorded counts were two or three. Stored-grid GLIF returned one spike in all five cells; recording-grid GLIF returned two for specimen 324493977 and one for the other four. In the other four Long Square prethreshold roles, recordings and both GLIF conditions contained zero spikes. Calculated Long Square GLIF steady-state errors for those supported rows ranged from 1.198 to 4.556 mV, while evaluation-cell conditional latency values ranged from 1.258 to 14.119 ms. These are named-case observations, not a cross-cell estimate, equivalence test, or model-condition ranking.

Case-specific GLIF spike counts in five evaluation cells

Marker positions use log_{1p} spacing; raw integer counts remain in the source table.

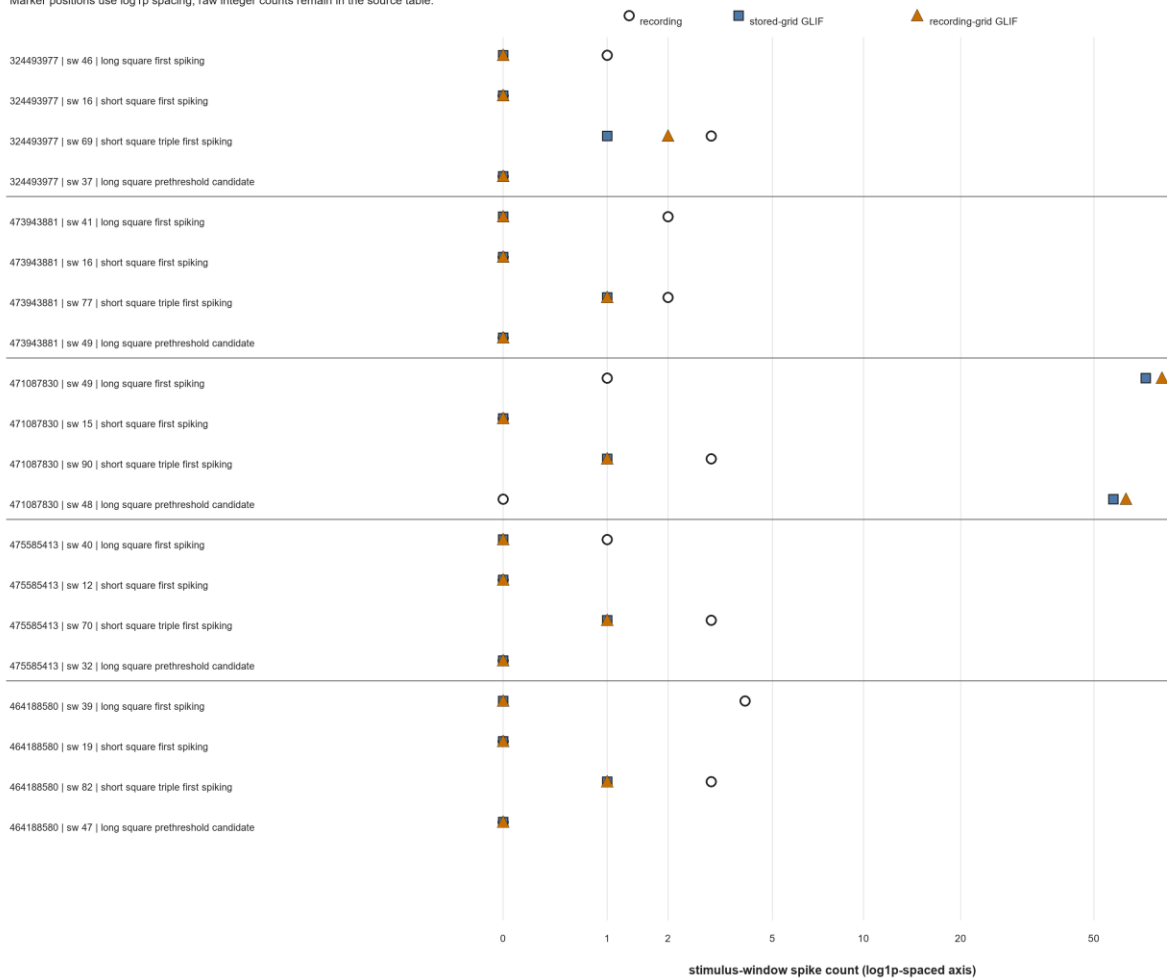


Figure 3. Case-specific GLIF spike counts in the five evaluation specimens. Recording, stored-grid GLIF, and recording-grid GLIF counts are shown for each role. Horizontal placement uses log_{1p} spacing so both low counts and the high-count specimen 471087830 cases remain visible; exact integers are in Supplementary Table S4. Marker shape and text, not color alone, identify the conditions.

The strict Short Square window excluded five near-offset thresholds

For every evaluation Short Square first-spiking trace, the prespecified half-open command interval [*onset*, *offset*) contained zero retained thresholds. A post hoc sensitivity extended the right boundary by 0.25 ms without changing the detector. It contained one retained threshold in each trace, occurring 0.040–0.200 ms after the prespecified offset. The filtered depolarizing upstroke was already engaged at the offset in all five traces (Figure 4). The extension did not replace the primary estimand, and no model metric was recalculated under it.

Short Square boundary sensitivity

Strict half-open interval is primary; +0.25 ms is post hoc.

Specimen / sweep	Strict primary count	+0.25 ms count	Threshold after offset	Upstroke at offset
324493977 / 16	0	1	0.145 ms	YES
473943881 / 16	0	1	0.095 ms	YES
471087830 / 15	0	1	0.040 ms	YES
475585413 / 12	0	1	0.200 ms	YES
464188580 / 19	0	1	0.200 ms	YES

The sensitivity does not replace the prespecified estimand or establish a protocol-general causal rule.

Figure 4. Short Square boundary sensitivity. Strict primary and post hoc sensitivity counts are shown in prespecified specimen order. The result is specific to these five traces, the prespecified event detector, and the stated boundary rule; it does not establish a protocol-general causal window.

Model construction and file-specific recording units were verified

The six perisomatic constructions were tied to exact fit JSON, SWC morphology, compiled mechanism, AllenSDK 2.16.2, and NEURON 8.2.7 identities. The construction analysis examined 158 fit rules. Every selector mapped nonemptily and exclusively to the intended normalized constructed section class; there were zero empty mappings, zero overinclusive mappings, and zero anomalies. The section inventory used normalized `h.allsec()` names because named HOC SectionLists were empty, while model assignments used direct selectors without fallback. The method-development cell had apical sections and rules; the other five evaluation-cell constructions did not. These results establish construction mapping identity, not morphology completeness, retention of disconnected source-SWC components, fit optimality, equilibrium, or physiological validity.

The file-specific unit analysis established response in volts and stimulus in amperes for all six NWB files. Five files used declared factor-1 conversion. The specimen 314822529 NWB was a legacy IVSCC pipeline-1.0 exception: defective `0.001` response and `1e-12` stimulus conversion attributes were bypassed under the AllenSDK pre-1.1 identity rule, with stored numerical values treated as SI. Display transformations alone converted volts to millivolts and amperes to picoamperes. The exception is file-specific and does not support a generic legacy conversion rule.

No original playback method satisfied all five synthetic benchmark waveforms

The delivered-current analysis tested the original two-argument `Vector.play` route and four alternatives against five prespecified synthetic benchmark waveforms at fixed $dt = 0.025$ ms (Table 3; Figure 5). The original route passed the one-source-sample-offset and midpoint benchmarks but failed the exact-boundary, sub-step-pulse, and consecutive-transition benchmarks. Explicit continuous playback and manual boundary sampling passed the first three benchmarks but failed sub-step pulses and consecutive transitions. Interval charge averaging passed only the exact-boundary benchmark under the joint transition, pulse, amplitude, and charge criteria. No candidate passed all five; consequently, none was applied to the 24 perisomatic cases.

Table 3. Delivered-current validation benchmark. “Criterion met” and “criterion not met” refer to the complete prespecified acceptance criteria for each synthetic waveform.

Candidate	Exact boundary	One-source offset	Midpoint	Sub-step pulse	Consecutive transitions	All five
Original two-argument <code>Vector.play</code>	Not met	Met	Met	Not met	Not met	No
Explicit time vector, discrete	Not met	Met	Met	Not met	Not met	No
Explicit time vector, continuous	Met	Met	Met	Not met	Not met	No
Manual boundary sample	Met	Met	Met	Not met	Not met	No
Solver-interval charge average	Met	Not met	Not met	Not met	Not met	No

Delivered-current validation benchmark

No tested method met the criteria for all five prespecified waveforms.

	exact boundary	one-source offset	midpoint	sub-step pulse	consecutive transitions
Original two-argument Vector.play	NOT MET	MET	MET	NOT MET	NOT MET
Explicit time vector, discrete	NOT MET	MET	MET	NOT MET	NOT MET
Explicit time vector, continuous	MET	MET	MET	NOT MET	NOT MET
Manual boundary sample	MET	MET	MET	NOT MET	NOT MET
Solver-interval charge average	MET	NOT MET	NOT MET	NOT MET	NOT MET

The 0.005 ms source grid can contain multiple transitions within one 0.025 ms solver interval.

A changed solver step or delivery rule would define a different analysis.

Figure 5. Delivered-current validation benchmark. Text and fill jointly indicate whether each method met the criteria for each waveform. A 0.005 ms source grid can contain multiple distinct transitions within one 0.025 ms solver interval. A single boundary value cannot expose all transitions, while interval averaging changes boundary amplitude. Changing the solver grid or the estimand would define a different analysis.

Post hoc reference implementations provide constructive delivery targets

We added a data-independent, piecewise-constant reference implementation after completing the original analysis. Fixed-step boundary sampling was evaluated at 0.025, 0.0125, 0.005, and 0.0025 ms using the unchanged waveform criteria. The two coarser steps met the complete criteria for three of five waveforms. The 0.005 and 0.0025 ms steps met the complete criteria for all five. An event-timed piecewise-constant reference also met all five criteria. These post hoc results establish constructive numerical targets and show that the benchmark can distinguish adequate temporal resolution from inadequate resolution. They do not establish compatibility with a particular NEURON current-injection mechanism, and no amended perisomatic analysis was run.

Perisomatic outputs met material-change criteria in all six cells

The initialization/equilibration analysis preserved the existing playback path so that sensitivity could be isolated from a delivery-method change. It used 48 independent simulations: one unchanged original-duration replay and one separately labeled extended-equilibration run for each of the 24 prespecified perisomatic cases. All reference time and voltage arrays exactly reproduced the original outputs. Nevertheless, every original pre-stimulus period

failed at least one prespecified voltage, state, or current stability criterion. Every sensitivity met the two-consecutive-window convergence rule at 3.0 s.

Twenty of 24 cases met at least one prespecified material-change rule (Table 4; Figure 6). All six cells therefore had at least three affected recording selections. Eleven reference cases contained stimulus-window spikes, whereas every sensitivity contained zero. Across the 12 Long Square cases with a defined comparison, steady-state shifts ranged from 9.482 to 25.440 mV. The four cases that did not meet a material-change threshold were Short Square zero/zero-spike cases; their original pre-stimulus periods were still drifting.

Table 4. Initialization/equilibration findings by cell. Each cell contributed four cases.

Specimen	Original pre-stimulus periods with drift	Extended equilibration converged	Material-change criterion met	Summary across recording selections
314822529	4/4	4/4	3/4	Criterion met in 3/4 selections
324493977	4/4	4/4	3/4	Criterion met in 3/4 selections
473943881	4/4	4/4	4/4	Criterion met in 4/4 selections
471087830	4/4	4/4	3/4	Criterion met in 3/4 selections
475585413	4/4	4/4	4/4	Criterion met in 4/4 selections
464188580	4/4	4/4	3/4	Criterion met in 3/4 selections

Initialization/equilibration sensitivity

All 24 original pre-stimulus periods drifted; all 24 extended periods converged.



Figure 6. Initialization/equilibration sensitivity by cell. Each block is one of four roles. “M” denotes a case meeting a prespecified material-change rule; “N” denotes a case that did not. Every original pre-stimulus period showed drift and every 3.0 s extended-equilibration run converged.

Independent replay recalculated all 576 retained trajectory slope/span statistics. The maximum absolute OLS-slope difference was $2.0328790734103208 \times 10^{-20}$, and the maximum span difference was zero. This confirms the reported calculations. It does not make the sensitivity a replacement primary analysis, demonstrate corrected physiology, or show that initialization alone explains model/recording differences. The delivered-current limitation remains independent.

Discussion

The results distinguish source identity, deterministic execution, and physiological interpretability. The six source files and model constructions could be identified precisely; prior outputs could be reproduced deterministically; and the analyses retained their prespecified event definitions and outcome-eligibility states. Yet no original delivered-current method satisfied all synthetic benchmark waveforms, and equilibration changed the unchanged perisomatic implementation under at least one material-change criterion in all six cells. File identity and repeatability were therefore insufficient to support a physiological interpretation of the perisomatic results.

The delivered-current result is not merely a missing-methods problem. It exposes a numerical incompatibility between a 0.005 ms source grid and a 0.025 ms fixed solver interval under the prespecified demand to preserve transitions, pulse widths, boundary amplitudes, and charge. Boundary sampling discards within-step structure; charge averaging preserves one quantity while altering others. The post hoc reference implementation shows that 0.005 ms fixed-step boundary sampling and exact event-timed piecewise-constant delivery can meet the same synthetic criteria. Translating either reference into a verified NEURON adapter and rerunning the perisomatic cases remains separate from the original analysis and cannot retroactively reinterpret it.

The equilibration result was limited to the tested conditions. The reference replay was exact, so the observed drift was not introduced by a changed computational path. The 3.0 s sensitivity converged by the prespecified rule and altered 20 cases, including spike outputs and Long Square steady state. This demonstrates dependence on initialization/equilibration under the unchanged implementation. It does not identify a biologically correct starting state, and because the same delivery path was retained, it cannot resolve the independent current-delivery problem.

The Short Square analysis illustrates a related estimand issue. The strict half-open interval was consistently applied and remained primary, but all five thresholds occurred just after its right boundary while the upstroke was already underway. Reporting both the prespecified result and the post hoc sensitivity is more informative than silently shifting the window or treating zero count error as general waveform agreement.

The GLIF rows show why case-specific reporting was preferable to a leaderboard. Several cells had similar stored- and recording-grid counts, but specimen 471087830 produced high-count Long Square outputs under both conditions and coverage-induced steady-state nulls. The paired conditions change both the command grid and the numerical step; their contrast is a coupled input-grid and model-dt sensitivity. Without known fitting-sweep exposure, a powered sample, or a common model-class estimand, a cross-cell average or preferred-condition claim would overstate the design.

The software and machine-readable reports provide explicit numerical checks alongside established approaches to model sharing and interoperability [10–12]. Their purpose is narrower than a universal simulator standard: they can accompany an executable neuronal model and reveal whether stimulus timing, state preparation, and outcome eligibility support the intended analysis.

Limitations

Six operationally selected cells are not representative of mouse cortex, and the method-development specimen is not part of the five-specimen evaluation set. Exposure of individual sweeps during model fitting remains unknown. The GLIF comparison jointly changes the input sampling grid and model time step and therefore cannot isolate either factor. The stimulus-delivery and initialization findings are specific to the tested implementations, versions, and criteria; the study does not identify a biologically correct initial state. Third-party payloads are retrieved by identifier rather than redistributed, and complete clean-machine reproduction of the full Allen-data workflow has not yet been demonstrated. The constructive playback extension is a model-free reference implementation; it has not been verified as a NEURON injection route or applied to the 24 perisomatic cases.

Methodological contribution

The study provides a current-delivery benchmark, an equilibration assessment, explicit reason-coded outcome-eligibility rules, and a software and provenance package that other groups can apply before interpreting neuronal model replay. Together these components distinguish repeatability from stimulus-delivery fidelity, model-state stability, and physiological interpretation without requiring population-level claims from the six-cell application.

Methods

Source data, animals, and ethics

Electrophysiology recordings, specimen metadata, GLIF configurations, perisomatic fit parameters, morphologies, and model identifiers were resolved from the Allen Cell Types resource [5]. Official Allen API v2 specimen queries were executed on 21 July 2026 for the six exact specimen IDs and recorded with query URLs and file digests. They established six distinct donors, ages P53–P63, four male and two female mice, genotypes, hemispheres, source structures, ephys-result IDs, and recording Well Known File (WKF) IDs. No new animals, tissue collection, or experimental procedures were used in the present secondary computational analysis. The source experimental procedures were reported as conducted under the Allen Institute Institutional Animal Care and Use Committee in the source study [7].

Analytical strata and prespecified roles

Specimen 314822529 was retained as a previously characterized method-development cell and was not treated as a sixth evaluation unit. The five evaluation cells were specimens 324493977, 473943881, 471087830, 475585413, and 464188580. Cell and sweep selection used metadata before response analysis. Selection chronology was fully reconstructable for the final acquisition group and only partially reconstructable for earlier groups.

The method-development cell retained Long Square (sweep 44), Short Square (17), Short Square Triple (75), and Square 2 s Suprathreshold (66) raw-label roles. Evaluation roles/sweeps were: 324493977, 46/16/69/37; 473943881, 41/16/77/49; 471087830, 49/15/90/48; 475585413, 40/12/70/32; and 464188580, 39/19/82/47, ordered as Long Square first-spiking, Short Square first-spiking, Short Square triple first-spiking, and Long Square prethreshold candidate. Cells were the analysis units; sweeps and model conditions were nested technical observations. The sample size was operational and no power calculation or inferential population model was used.

Recording units and command representation

Each recording was linked to an exact WKF ID and file digest. Five NWBs exposed response in volts and stimulus in amperes through factor-1 conversion. For exact recording WKF 491202285, specimen 314822529, defective legacy conversion attributes were bypassed under the AllenSDK pre-1.1 compatibility branch. Stored numerical SI values were used. Display-only transformations were $\text{mV} = V \times 1,000$ and $\text{pA} = A \times 10^{12}$.

Recording-grid commands were represented at 200 kHz. Stored-grid GLIF commands used nonoverlapping factor-10 arithmetic means and block-start times to produce 20 kHz inputs without amplitude scaling, interpolation, or padding. Perisomatic source commands were converted from amperes to nanoamperes by `stimulus_nA = stimulus_A × 109` before fixed-step playback. This conversion specifies the source vector; it does not establish solver-time delivered-current fidelity.

Model conditions and exact construction

The 24 recording selections were executed or represented in three separate simulation conditions. GLIF stored-grid used `dt = 50 μs` at 20 kHz; GLIF recording-grid used `dt = 5 μs` at 200 kHz. Stored GLIF model spike times were taken from the prespecified AllenSDK-returned spike arrays rather than redetected from model voltage. GLIF models followed the Allen generalized leaky integrate-and-fire framework [1].

Perisomatic construction used specimen-specific fit JSON, SWC morphology, and compiled mechanisms through AllenSDK 2.16.2 with NEURON 8.2.7 [2–4]. The construction route included AllenSDK’s soma-attached two-section axon replacement and HOC %g serialization of section-targeted values. The construction analysis instantiated models only, did not call `finalize`, dispatch a stimulus, or advance simulation time, and verified every fit selector against normalized constructed-section names.

The perisomatic reference execution used fixed `dt = 0.025 ms`, inactive CNode, the documented two-argument `Vector.play` route, and `h.initialize()` at `t = 0`. These settings define execution provenance. The delivered-current and equilibration methods below separately tested delivery and pre-stimulus behavior.

Event detection, windows, and masking

Recorded spikes followed the prespecified AllenSDK 2.16.2-equivalent detector: full-trace detection, a four-pole zero-phase Bessel low-pass filter at 10 kHz, positive derivative cutoff 20 V/s, minimum peak –30 mV, minimum threshold-to-peak height 2 mV, maximum threshold-to-peak interval 5 ms, threshold refinement at 0.05 of mean

retained upstroke, and final-return tolerance of threshold plus 1 mV. No fixed refractory period was imposed. Overlapping threshold/peak pairs were merged under the prespecified negative-derivative rule.

Stimulus windows were half-open. Each recording or model threshold created a half-open mask [$t - 1 \text{ ms}$, $t + 5 \text{ ms}$). Recording and model intervals were unioned and independently projected to each native grid by physical-time membership, without trace interpolation. Source-specific Long Square baseline and steady-state windows were retained from the prespecified case definitions rather than recomputed after inspecting the combined results.

Outcomes and analysis

Absolute spike-count error was $|\text{model count} - \text{recording count}|$ and applied to every case. Conditional absolute first-spike-latency error was $|\text{model first spike} - \text{recording first spike}| \times 1,000 \text{ ms}$ only when both traces contained at least one retained in-window event. Long-duration absolute steady-state delta-voltage error was the absolute difference between model and recording baseline-referenced steady-state changes, expressed in mV. It applied to Long Square and the method-development Square - 2s Suprathreshold label and required at least 0.80 usable unmasked coverage independently in recording and model baseline and steady-state windows. Unavailable metrics retained the exact null reason and missingness class.

Analysis was descriptive and case-level. No composite score, correlation, full-trace RMSE, normalized score, arm-average, cross-cell mean, confidence interval, or hypothesis test was part of the prespecified analysis. No ranking or selection was performed. All displayed rounding is presentational; full-precision values remain in the machine-readable tables.

Short Square boundary sensitivity

For each primary Short Square first-spiking trace, the detector was replayed over the full corrected trajectory. Counts were evaluated under the strict [command_onset , command_offset) window and a separately labeled [command_onset , $\text{command_offset} + 0.25 \text{ ms}$) sensitivity. Threshold/peak timing, command indices, voltage, and filtered derivative values at the prespecified offset were retained. The strict window remained the primary estimand.

Delivered-current validation benchmark

Five model-free synthetic benchmark waveforms tested exact-grid boundaries, a one-source-sample offset, a midpoint transition, a sub-solver-step pulse, and consecutive transitions. Five candidate routes were assessed: original two-argument vector playback; explicit time-vector discrete and continuous playback; manual boundary sampling; and solver-interval charge averaging. Acceptance jointly evaluated transition count, pulse structure, boundary amplitude, and interval-aware charge against prespecified tolerances. Satisfying all waveform criteria was required before applying a candidate method to the 24 perisomatic cases. The analysis used Python 3.11.15 and NEURON 8.2.7 in the specified WSL environment.

Post hoc constructive playback extensions

After completing the original five-method analysis, we implemented a model-free piecewise-constant reference in Python. Fixed-step boundary sampling used solver steps of 0.025, 0.0125, 0.005, and 0.0025 ms. At each solver boundary, it selected the zero-order-held source value without interpolation. A separate event-timed reference applied every source transition at its specified time. Both extensions were evaluated using the original transition-count, transition-timing, pulse-width, boundary-amplitude, and interval-charge criteria and tolerances. The benchmark reports full criterion-level results in deterministic JSON. Run-specific timing and interpreter metadata are omitted because they are not scientific outcomes and do not affect the model-free calculations.

Initialization/equilibration sensitivity

For each of 24 prespecified perisomatic cases, one unchanged reference process and one sensitivity process were executed. Reference runs had to reproduce the original time and voltage arrays exactly. Pre-stimulus telemetry included prespecified voltage, gating-state, and available membrane/current trajectories. Stability was assessed in final 100 ms windows using prespecified slope/span and current/state criteria. Sensitivities advanced the unstimulated model until two consecutive windows passed, with convergence capped by the protocol; all converged at 3.0 s.

Material change followed prespecified rules for stimulus-window spike count, event timing, coverage, and steady-state voltage. Sensitivity outputs remained separate from reference rows. An independent calculation reproduced 576 trajectory statistics and separately evaluated the two-window rule. Only runs generated with the final recorder-binding and data-capture configuration were analyzed.

Reproducibility

The combined analysis reconciled six cells, 24 roles, 72 cases, and 216 outcome-applicability entries. Repeating the delivered-current analysis in the specified scientific environment produced byte-identical result files. Repeating the equilibration analysis reproduced all 24 reference trajectories, all 24 convergence classifications, and all retained trajectory statistics. File digests were used to verify source and result identity. Automated tests for the analysis code and tabular consistency are provided with the public software.

Software environment

The scientific environment used micromamba 2.8.1, Python 3.11.15, NEURON 8.2.7 (commit [34cf696c4](#)), AllenSDK 2.16.2, NumPy 1.23.5, GCC 14.3.0, Ubuntu 26.04 LTS, GLIBC 2.43, and Linux kernel 6.18.33.1-microsoft-standard-WSL2.

Data and code availability

Study code, derived tables, figures, and machine-readable provenance are available in the public distribution under the MIT license. The accompanying data identify specimens, recordings, models, fits, morphologies, mechanisms, and source URLs. Allen NWB files, model data, and compiled third-party binaries are not

redistributed; they must be retrieved from their original sources using the identifiers provided. Complete reproduction from a fresh public environment has not yet been demonstrated [8].

References

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